Extended Named Entity Recognition Using Finite-State Transducers

***Mauro Gaio*[1], *Ludovic Moncla*[1]**
[1] *Université de Pau et des Pays de l'Adour, LIUPPA, France*
{mauro.gaio,ludovic.moncla}@univ-pau.fr

Introduction

Context

The PERDIDO project

- Project for Extracting and Retrieving Displacements from textual Documents

`http://erig.univ-pau.fr/PERDIDO/`

Wider context

- Digital humanities :
  - Enhancement of cultural heritage : travelogues
  - Tourism : hikes, treks, etc.
  - Analysis of population migration

Introduction

A Pralognan suivre la route entre l'hôtel de la Vanoise et celui du Petit Mont Blanc et continuer tout droit. Passer les bourgs de Barioz et de Bieux. On tombe alors sur le GR55 à suivre jusqu'au hameau des Fontanettes. Le sentier suit par la gauche un télésiège jusqu'au refuge des Barmettes. Poursuivre par le pont de la Glière. Arrivé près des chalets de la Glière, ne pas prendre le sentier de droite qui mène au Moriond mais filer tout droit. Plus loin passer sur le pont du Chanton. Peu après, on parvient au lac des Vaches que l'on traverse sur des rochers. Le dénivelé durant cette partie est assez conséquent, de l'ordre de 1100m, mais rien que le
méritent le détour. Au bout
carrefour, ne pas partir ver
direction du lac Long que l'on
refuge du Col de la Vanoise.
dénivelé contourne le Lac Ro
une bonne descente de 300
pont de Cro
hauteur, au efour, ne p
partir à gau t rester su
pendant plus kilomètre
vallée sauvag ncaissée
douce mais rég Au loin
peu dure à gravir fin

Introduction

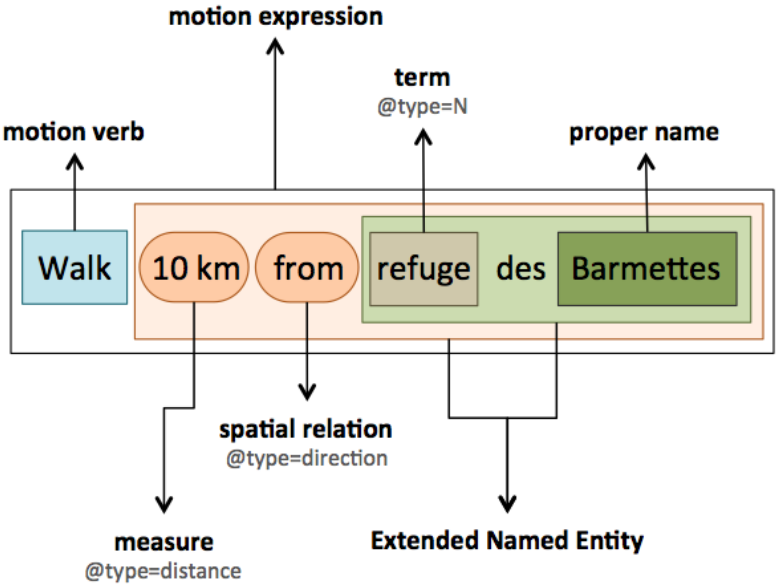
Background

Space and Motion in Language

- Space in language (Talmy, 1985 ; Vandeloise, 1986 ; Aurnague and Vieu, 2015)
  - Reference object (*ground*, *site*)
  - Object to be located (*figure*, *target*)
  - Spatial relations between them
- Classification of verbs (Boons, 1987 ; Laur, 1993 ; Muller, 1998)
  - Polarity : initial (*to leave*), median (*to cross*), final (*to reach*)
  - Prepositions (*to*, *from*, . . .)
- Named Entities
  - Typology : persons, locations, organizations, . . . (Tran, 2006 ; Ehrmann 2008)
  - Recognition and classification tasks (Nadeau and Sekine, 2007 ; Friburger and Maurel, 2004)

Annotating Spatial Descriptions

Overview



Annotating Spatial Descriptions

Extended Named Entity (ENE)

Two categories of proper names

- pure : proper names only (simple or complex)
- descriptive : composition of proper names and common nouns

Descriptive proper names

- NE built with a pure proper name and descriptive expansion
- expansion can change the implicit type (location, person, etc.)

Extended Named Entity

- composed of pure or descriptive proper names
- several levels of overlapping

Annotating Spatial Descriptions

Extended Named Entity (ENE)

Level 0 (pure proper name)

(1) a. *Nice* → one entity (location)

b. *Greenpeace* → one entity (organisation)

c. *Charles de Gaulle* → one entity (person)

Level 1 (descriptive proper name) : same types

- descriptive expansions may not change the implicit or default nature of the object described by the proper name

(2) *comunidad autónoma de* *Aragón*
'autonomous community of Aragón'

Annotating Spatial Descriptions

Extended Named Entity (ENE)

Level 1 : different types

- descriptive expansions may change the implicit or default nature of the object described by the proper name

(3) *maire de* $\boxed{\textit{Nice}}$

'mayor of Nice'

→ two entities, ***Nice*** (location) and ***maire de Nice*** (person)

Level > 1

(4) *portavoce della* $\boxed{\textit{Villa Médicis a } \boxed{\textit{Roma}}}$

'spokesperson of the Villa Médicis in Rome'

Annotating Spatial Descriptions

Extended Named Entity (ENE)

Geoparser based on construction grammars

- Concept of construction as a theoretical entity
- A construction is a pattern used to
  - generate the elements of the language,
  - extract these elements.

Annotating Spatial Descriptions

Extended Named Entity (ENE)

- ENER
  - level: 3
  - type: *place name*
  - comp.: NP,OFFSET,
    - ENER
      - level: 2
      - type: *place name*
      - comp.: NP,IN,OFFSET,
        - ENEA
          - level: 1
          - type: *location*
          - cat.: *descriptive*
          - comp.: NN,IN,
            - ENEA
              - level: 0
              - type: *location*
              - cat.: *pure*
              - comp.: NNP
              - lex.: *Aragon*
          - lex.: *region of Aragon*
      - lex.: *arid territory on the south of the region of Aragon*
  - lex.: *karst depression on the arid territory on the south of the region of Aragon*

Annotating Spatial Descriptions

Motion event expressions

Construction grammars VT

- mark and formalise the relations between ENE, geographical terms, spatial relations and movement verbs

(5) *Emprunter successivement **rue des Capucins** et **rue de Compostelle**.*
'Walk down **Capucins Street** and then **Compostelle Street**.'

(6) *Prendre à gauche après l'entrée de **l'usine de Fontanille**.*
'Turn left after the entry to the **Fontanille factory**.'

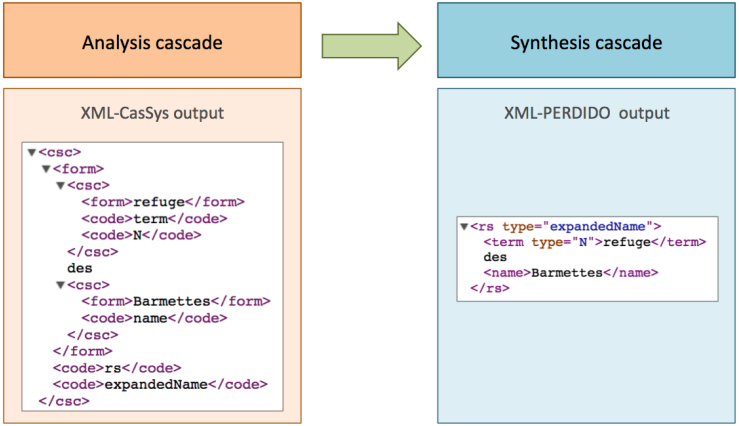
(7) *Suivre la route depuis le **hameau Lic** jusqu'à la **Chapelle Saint-Roche**.*
'Follow the road from the **hamlet Lic** to the **Chapelle Saint-Roche**.'

Annotating Spatial Descriptions

Cascaded finite-state transducers

Cascaded finite-state transducers

- CasSys program in the Unitex platform



Annotating Spatial Descriptions

Cascaded finite-state transducers

XML-TEI output format

```
<TEI xmlns="http://www.tei-c.org/ns/1.0">
  <text><body><p><s>
    <phr type="verb_phrase" subtype="motion">Walk
      <measure type="distance">10 km</measure>
      <offset type="direction" subtype="initial">from</term>
      <placeName n="1" ref="www.openstreetmap.org/node/451703419">
        <geogName type="S" subtype="RHSE">
          <geogFeat>refuge</geogFeat>des<name>Barmettes</name>
        </geogName>
      </placeName>
    </phr>
  </s></p></body></text>
</TEI>
```

Evaluation

Named Entity Recognition and Classification

Corpus

- Hiking descriptions
- French, Spanish and Italian

	ENE	*Perdido*	
level 0	304	244	80%
level 1	332	280	84%
level 2	20	17	85%
level 3	4	1	25%
total	**660**	**542**	**82%**

TABLE – Number of correctly detected ENE with Perdido (French)

Conclusions

Geoparser based on construction grammars

- cascaded finite-state transducers
- accessible through web services

Concept of ENE

- local context associated with NE

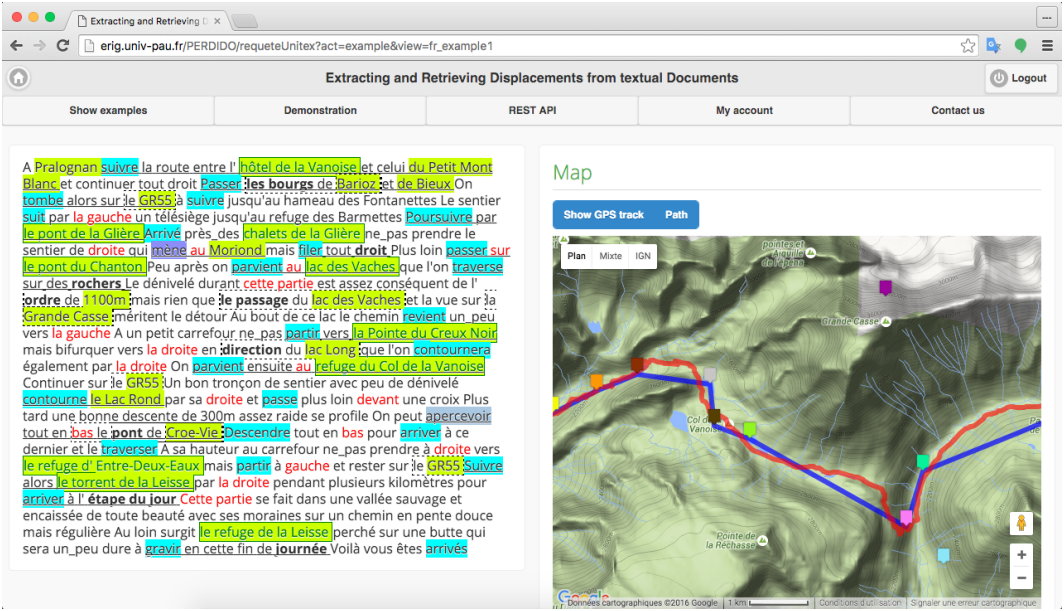
Construction grammars VT

- mark and formalise the relations between ENE, geographical terms, spatial relations and movement verbs

Web services

`http://erig.univ-pau.fr/PERDIDO/api.jsp`

Conclusions



Thank you for your attention

CONTACT

Mauro Gaio (LIUPPA)
mauro.gaio@univ-pau.fr

Ludovic Moncla (LIUPPA)
ludovic.moncla@univ-pau.fr

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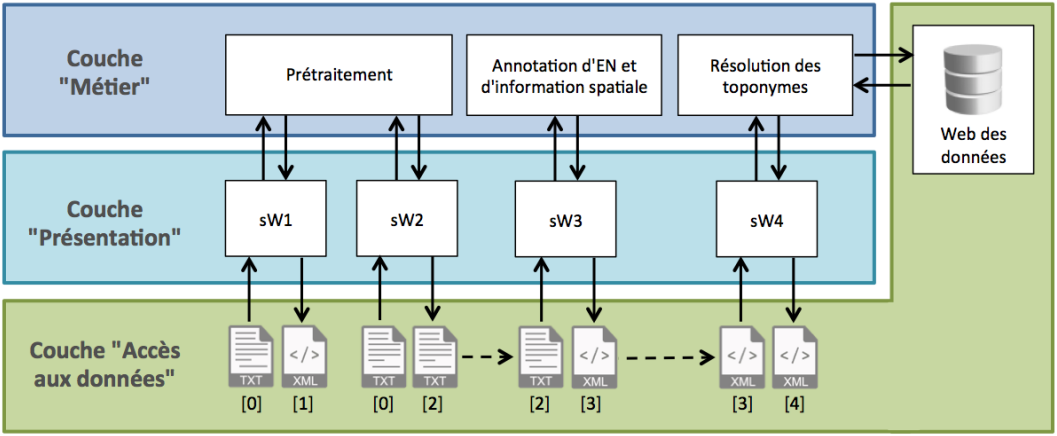
PPP Pau Porte des Pyrénées Communauté d'agglomération

Annotating Spatial Descriptions

Implementation



Comparison of the French POS taggers

Proper names

	TreeTagger	FreeLing	Talismane
Precision	94.3%	90.5%	94.3%
Recall	87.3 %	98.1%	96.0%
F1-Measure	90.6%	94.2%	95.1%

Verbs

	TreeTagger	FreeLing	Talismane
Precision	97.4%	96%	99.5%
Recall	99.8%	99.8%	99.5%
F1-Measure	98.6%	97.8%	99.5%

Evaluation

Slot Error Rate (SER) :

- insertions (I)
- deletion (D)
- substitution
  - classification (C)
  - boundaries detection (B)
  - and both (CB)

$$SER = \frac{I + D + 0.5C + 0.5B + CB}{relevantresults}$$

	level 0	level 1	level 2	level 3	**total**
SER	31.1%	13%	7.5%	37.5%	**16.7%**
Recall	98%	93.7%	100%	100%	**95.9%**
Precision	90.9%	98.7%	100%	100%	**94.9%**

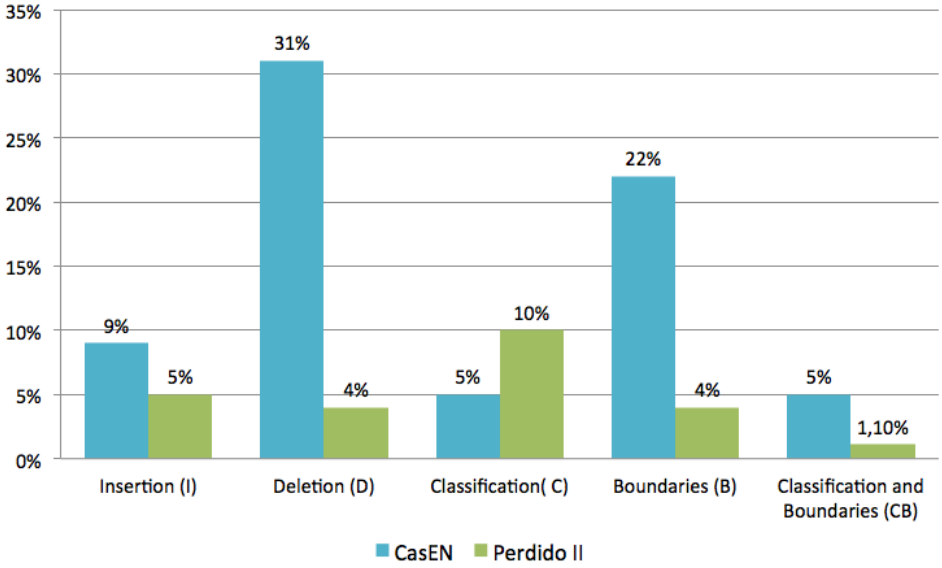
TABLE – Evaluation of the NERC task (French)

Gold-standard corpus : Distribution of verbs

	French		Spanish		Italian	
Total # of verbs	1694		867		805	
# of motion verbs	1101	(65%)	456	(53%)	428	(53%)
- initial	66	(6%)	74	(16%)	34	(8%)
- median	710	(64%)	216	(47%)	255	(60%)
- final	325	(30%)	166	(37%)	139	(32%)
# of perception verbs	41	(2%)	36	(4%)	36	(4%)
# of topographic verbs	51	(3%)	54	(6%)	49	(6%)

Gold-standard corpus : Distribution of verbs

French		Spanish		Italian	
prendre	188	llegar	64	proseguire	44
suivre	100	recorrer	34	seguire	41
traverser	78	seguir	34	raggiungere	29
arriver	71	pasar	31	arrivare	24
continuer	64	tomar	28	attraversare	22
descendre	61	continuar	27	salire	22
passer	60	visitar	21	continuare	20
monter	51	salir	20	scendere	18
rejoindre	44	dirigir	19	portare	18
partir	35	ir	17	percorrere	15


35%
30%
25%
20%
15%
10%
5%
0%
31%
22%
9%
5%
4%
5%
10%
4%
5%
1,10%
Insertion (I)
Deletion (D)
Classification(C)
Boundaries (B)
Classification and Boundaries (CB)
CasEN
Perdido II

Annotating Spatial Descriptions

Extented Named Entity (ENE)

The core of the grammar

- bottom-up strategy
- two types of ENE :
  - absolute
  - relative

$S \rightarrow ENE$
$ENE \rightarrow ENEA \quad | \quad (Term)\ ENER$
$ENER \rightarrow Offset\ ENEA \quad | \quad Offset\ ENER$
$ENEA \rightarrow (Term)\ ProperNoun \quad | \quad Term\ ENEA$
$Term \rightarrow Nominal\ Det$
$Nominal \rightarrow Noun \quad | \quad Nominal\ Noun$

Annotating Spatial Description

Motion event expressions

Construction grammars VT

- mark and formalise the relations between ENE, geographical terms, spatial relations and movement verbs

$$
\begin{array}{l}
S \rightarrow V\ T \\
V \rightarrow Verb \quad | \quad Verb\ SO \\
C \rightarrow Conjonction \mid , \\
LT \rightarrow ENE\ C\ T \\
T \rightarrow (SO)\ (det)\ ENE \quad | \quad (SO \quad | \quad ENE)\ T \quad | \quad (SO)\ LT
\end{array}
$$

- V a set of movement verbs
- T a set of n-tuples composing of :
  - SO a set of spatial offset
  - TG a set of geographical noun phrase
  - E a set of ENE